Anthony Garcia

Computer Science - B.Sc. | Toronto Metropolitan University
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Education

• **Toronto Metropolitan University (Toronto, ON)** *09/2023 - Expected 2028*
Bachelor of Science in Computer Science cGPA: 4.19 / 4.33

Personal Projects

• **Robotic Arm Pick-and-Place with Reinforcement Learning** *garciaa047/PandaPlaceRL*
MuJoCo environment where a Franka Panda arm learns pick-and-place via SAC + HER through curriculum learning.

- Tools & Technologies: Python, MuJoCo, Gymnasium, Stable-Baselines3, PyTorch, TensorBoard
- Designed a custom goal conditioned Gymnasium environment with 4D Cartesian control via damped least squares Jacobian inverse kinematics, sparse HER compatible rewards, and optional shaping terms.
- Achieved 96% to 100% success across three evaluation suites with fully randomized object and goal positions, converging in roughly 400k steps (about 1.5 hours on CPU).
- Engineered a six stage performance gated curriculum with mixed goal sampling and grasped starts, enabling sparse reward learning of grasping behaviours that random exploration never discovers.

• **Neural Network Image Classifier from Scratch** *garciaa047/Neural-Network-Doodle-Recognition-System*
Built a feedforward neural network from scratch in NumPy achieving 96% test accuracy on 5 class doodle recognition.

- Tools & Technologies: Python, NumPy, Pillow, Tkinter
- Engineered a neural network (900→128→64→5) with manual backpropagation, Leaky ReLU, He initialization, softmax output, L2 regularization, momentum, and mini batch gradient descent, with no ML frameworks.
- Built a custom data processing pipeline sampling 3,000 images per class, splitting train/test before augmentation and using validation based model selection to avoid keeping overfit epochs, reaching 94.5 to 97.7% accuracy on every held out class.
- Designed a data augmentation pipeline (rotation, translation, Gaussian noise) tripling training data, and built a real time Tkinter drawing interface with live classification at ∼75ms prediction intervals.

Experience

• **Royal Bank of Canada (RBC)** *05/2026 - 12/2026*
Software Developer Toronto, ON

- Developed and shipped full stack features for RBC's client facing mortgage pre-approval platform, serving tens of thousands of clients and facilitating millions of dollars in pre-approvals.
- Built responsive front end components in Angular (TypeScript) with an in house framework and implemented backend REST services in Java Spring Boot with SQL data access.
- Created Python and SQL scripts and internal developer tools, and designed team workflows that streamlined development processes across the squad.

• **Bimen Business Consultants** *05/2025 - 08/2025*
Project Coordinator Toronto, ON

- Developed a PowerShell automation script for a client to sort and file 500+ legal documents daily, eliminating manual processing and implementing email notifications with error reporting.
- Performed quality assurance testing for a client software rollout, documenting bugs and providing UX improvement recommendations that influenced multiple feature updates.
- Supported technical infrastructure migration during a client's ownership transition, including server provisioning, data migration, and user account creation for employees.

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL
Frameworks & Libraries: Spring Boot, Angular, MuJoCo, Gymnasium, Stable-Baselines3, NumPy, Pandas
Tools & Concepts: Git, REST APIs, Agile/Scrum, Reinforcement Learning, Machine Learning, Imitation Learning (Behavior cloning), AI, Research, Neural Networks, Data Structures, Algorithms, Object Oriented Programming
Coursework: Data Structures, Algorithms, Databases, Operating Systems, Computer Security

Achievements

• **TechPowHer Hackathon - 2nd Place Overall** of 22 teams, 178 participants *07/2026*

• **Academic Excellence Renewable Scholarship ($3000 x 3 = $9000 Total)** *Requires: GPA 4.15+*